

Chapter 6

Random Number

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Random Number

Random numbers are samples drawn from a uniformly distributed random variable between some satisfied intervals, they have equal probability of occurrence.

6.1 Properties of Random Number

- Properties of random number has two important statistical properties.
 1. Uniformity and
 2. Independence

6.1 Properties of Random Number

Each random number R_t is an independent sample drawn from a continuous uniform distribution between 0 and 1

$$\text{pdf: } f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

6.1 Properties of Random Number

If the interval between 0 and 1 is divided into n equal parts or classes of equal length, then

- The probability of observing a value in a specified interval is independent of previous value drawn.
- If a total of m observations are taken, then the expected number of observations in each interval is m/n , for uniform distribution.

Pseudo Random Numbers

- The pseudo means false.
- Pseudo implies that the random numbers are generated by using some known arithmetic operation.
- Since, the arithmetic operation is known and the sequence of random numbers can be repeatedly obtained, the numbers cannot be called truly random.
- However, the pseudo random numbers generated by many computer routines , very closely fulfill the requirement of desired randomness.

Pseudo Random Numbers

- If the method of random number generation that is the random number generator is defective, the generated pseudo random numbers may have following departures from ideal randomness.
 - The generated numbers may not be uniformly distributed.
 - The generated numbers may not be continuous.
 - The mean of the generated numbers may be too high or too low.
 - The variance may be too high or too low.

Pseudo Random Numbers

- There may be cyclic patterns in the generated numbers, like;
 - a) Auto co-relation between numbers
 - b) a group of numbers continuously above the mean, followed by group continuously below of mean.
- Thus, before employing a pseudo random number generator, it should be properly validated, by testing the generated random numbers for randomness.

Pseudo Random Numbers

Generation of Random Number:

- Should be fast
- Should be portable
- Should have sufficiently long cycle
- Should be replicable
- Should hold ideal statistical properties (uniformity and independence)

6.2 Generation of pseudo-random numbers

- In computer simulation, where a very large number of random numbers is generally required, the random numbers can be obtained by the following methods.
 1. Random numbers may be drawn from the random number tables stored in the memory of the computer.
 2. Using electronics devices-Very expensive
 3. Using arithmetic operation

6.3 Random Number Generation

Linear Congruential Method or Mixed Congruential Method :

- The linear congruential method generates a sequence of integers $X_1, X_2, \dots$ Between zero and $m-1$ by following recursive relationship:
- $X_{i+1} = (a X_i + c) \bmod m, i = 0, 1, 2, \dots$ Where the initial value X_0 is called seed, a is the multiplier, c is the increment and m is the modulus.
- When $c = 0$, the form is known as multiplicative congruential method.
- Using this relation, the random number can be generated as:

$$R_i = X_i / m, i = 1, 2, 3, \dots$$

Linear Congruential Method or Mixed Congruential Method: Example

Example 1: let $X_0 = 27$, $a = 17$, $c = 43$, and $m = 100$, then $i = 5$.

$$X_1 = (17*27 + 43) \bmod 100 = 2$$

$$R_1 = 2 / 100 = 0.02$$

$$X_2 = (17*2 + 43) \bmod 100 = 77$$

$$R_2 = 77 / 100 = 0.77$$

$$X_3 = (17*77 + 43) \bmod 100 = 52$$

$$R_3 = 52 / 100 = 0.52$$

$$X_4 = (17*52 + 43) \bmod 100 = 27$$

$$R_4 = 27 / 100 = 0.27$$

$$X_5 = (17*27 + 43) \bmod 100 = 2$$

$$R_5 = 2 / 100 = 0.02$$

Therefore, the generated random numbers after 5 iterations are 0.02, 0.77, 0.52, 0.27 and 0.02.

Linear Congruential Method or Mixed Congruential Method: Example

- Example 2: Use Linear Congruential Method to generate a sequence of 5 random numbers. Let $X_0 = 29$, $a = 9$, $c = 49$, and $m = 100$.

Linear Congruential Method or Mixed Congruential Method: Example

- Example 2: Use Linear Congruential Method to generate a sequence of 5 random numbers. Let $X_0 = 29$, $a = 9$, $c = 49$, and $m = 100$.

$$R_0 = 29/100 = 0.29$$

$$X_1 = (9*29 + 49) \bmod 100 = 10$$

$$R_1 = 10/100 = 0.1$$

$$X_2 = (10*9 + 49) \bmod 100 = 39$$

$$R_2 = 39/100 = 0.39$$

$$X_3 = (39*9 + 49) \bmod 100 = 0$$

$$R_3 = 0/100 = 0$$

$$X_4 = (0*9 + 49) \bmod 100 = 49$$

$$R_4 = 49/100 = 0.49$$

6.3 Random Number Generation Methods

Multiplicative Congruential Method:

- Basic Relationship
- $X_{i+1} = (a X_i) \bmod m$, where $a \geq 0$ and $m \geq 0$, most natural choice for m is one that equals to the capacity of a computer word.
- $m = 2^b$ (binary machine), where b is the number of bits in the computer word.
- $m = 10^d$ (decimal machine), where d is the number of digits in the computer word.

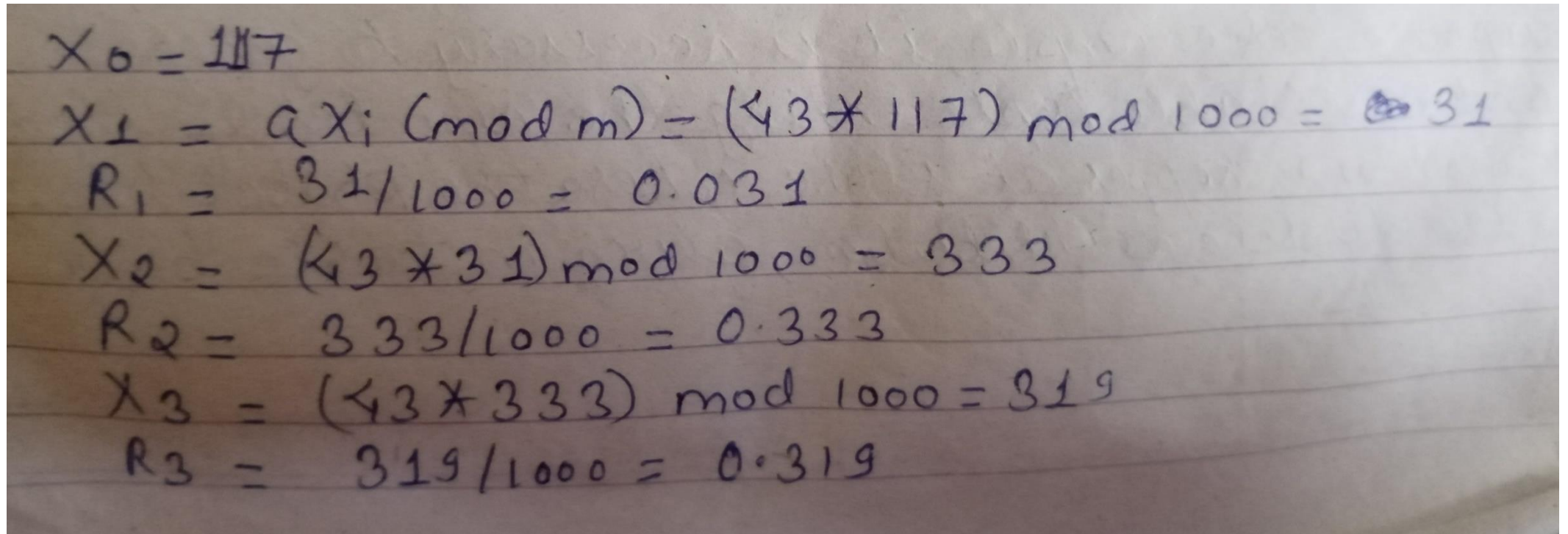
Multiplicative Congruential Method Example

- Example 1: Use Multiplicative Congruential Method to generate sequence of 3 random numbers with $X_0 = 117$, $a = 43$ and $m = 1000$.

Multiplicative Congruential Method Example

- Example 1: Use Multiplicative Congruential Method to generate sequence of 3 random numbers with $X_0 = 117$, $a = 43$ and $m = 1000$.

Solution,



Handwritten solution showing the steps to generate random numbers using the Multiplicative Congruential Method:

$$\begin{aligned} X_0 &= 117 \\ X_1 &= aX_0 \pmod{m} = (43 \times 117) \pmod{1000} = 31 \\ R_1 &= 31/1000 = 0.031 \\ X_2 &= (43 \times 31) \pmod{1000} = 333 \\ R_2 &= 333/1000 = 0.333 \\ X_3 &= (43 \times 333) \pmod{1000} = 319 \\ R_3 &= 319/1000 = 0.319 \end{aligned}$$

Therefore, the generated random numbers are 0.031, 0.333 and 0.319

Maximum Period Calculation

- The max period(P) is:
 - Case 1: For m a power of 2, say $m = 2^b$, and $c \neq 0$, the longest possible period is $P = m = 2^b$, which is achieved provided that c is relatively prime to m (that is, the greatest common factor of c and m is 1), and $a = 1 + 4k$, where k is an integer.
 - Case 2: For m a power of 2, say $m = 2^b$, and $c = 0$, the longest possible period is $P = m / 4 = 2^{b-2}$, which is achieved provided that the seed X_0 is odd and the multiplier, a , is given by $a = 3 + 8k$ or $a = 5 + 8k$, for some $k = 0, 1, \dots$
 - Case 3: For m a prime number and $c = 0$, the longest possible period is:
 $P = m - 1$, which is achieved provided that the multiplier, a , has the property that the smallest integer k such that $a^k - 1$ is divisible by m is $k = m - 1$.

Maximum Period Calculation Examples

Example 1 : Using the Linear congruential method, find the period of the generator for $a = 5$, $m = 4$, and $X_0 = 1$ and $c = 43$.

Solution,

Maximum Period Calculation Examples

Example 1 (Case 1): Using the Linear congruential method, find the period of the generator for $a = 5$, $m = 4$, and $X_0 = 1$ and $c = 43$.

Solution,

m can be expressed as $2^2 = 4$, $c = 43$ (non-zero) and is relatively prime to m , and 'a' satisfies $a = 1 + 4k$ when k is taken as 1 i.e. $5 = 1 + 4*1$

Hence, $P = m = 4$.

Maximum Period Calculation Examples

$$X_1 = (5 \cdot 1 + 43) \bmod 4 = 0$$

$$R_1 = 0 / 4 = 0$$

$$X_2 = (5 \cdot 0 + 43) \bmod 4 = 3$$

$$R_2 = 3 / 4 = 0.75$$

$$X_3 = (5 \cdot 3 + 43) \bmod 4 = 2$$

$$R_3 = 2 / 4 = 0.5$$

$$X_4 = (5 \cdot 2 + 43) \bmod 4 = 1$$

$$R_4 = 1 / 4 = 0.25$$

$$X_5 = (5 \cdot 1 + 43) \bmod 4 = 0$$

$$R_5 = 0 / 4 = 0$$

<u>i</u>	X_i	<u>R_i</u>
1	0	0
2	3	0.75
3	2	0.5
4	1	0.25
5	0	0

Maximum Period Calculation Examples

- Example 2: Using the multiplicative congruential method, find the period of the generator for $a = 13$, $m = 64$, and $X_0 = 1$, and 3.

Maximum Period Calculation Examples

- Example 2 (Case 2): Using the multiplicative congruential method, find the period of the generator for $a = 13$, $m = 64$, and $X_0 = 1$, and 3.
- Solution, m can be expressed as $2^6 = 64$, $X_0 = 1$ and 3; X_0 is odd and 'a' satisfies $a = 5 + 8k$ when $k = 1$ i.e. $13 = 5 + 8 \cdot 1$. Hence, period is:

$$\begin{aligned} P &= m/4 \\ &= 64/4 \\ &= 16 \end{aligned}$$

Multiplicative Congruential Method Examples

<u>i</u>	X_i	X_i
0	1	3
1	13	39
2	41	59
3	21	63
4	17	51
5	29	23
6	57	43
7	37	47
8	33	35
9	45	7
10	9	27
11	53	31
12	49	19
13	61	55
14	25	11
15	5	15
16	1	3

Multiplicative Congruential Method Examples

- Example 3 (Case 3): Using the multiplicative congruential method, find the period of the generator for $a = 2$, $m = 5$, and $X_0 = 1$ and $c = 0$.

Solution,

Here, $m = 5$ is a prime number and $c = 0$

Given that $m = 5$ we can get $k = m - 1 = 4$

When $a = 2$, we get $a^k - 1 = 2^4 - 1 = 16 - 1 = 15$ and 15 is exactly divisible by 5 hence period can be calculated as $P = m - 1$.

$$= 5 - 1$$

$$= 4$$

Multiplicative Congruential Method Examples

$$X_1 = (2 * 1) \bmod 5 = 2$$

$$R_1 = 2/5 = 0.4$$

$$X_2 = (2 * 2) \bmod 5 = 4$$

$$R_2 = 4/5 = 0.8$$

$$X_3 = (2 * 4) \bmod 5 = 3$$

$$R_3 = 3/5 = 0.6$$

$$X_4 = (2 * 3) \bmod 5 = 1$$

$$R_4 = 1/5 = 0.2$$

$$X_5 = (2 * 1) \bmod 5 = 2$$

$$R_5 = 2/5 = 0.4$$

i	X_i	R_i
1	2	0.4
2	4	0.8
3	3	0.6
4	1	0.2
5	2	0.4

Summary of requirements for a good pseudorandom generator

1. The sequence of random numbers generated must follow the uniform $(0, 1)$ distribution.
2. The sequence of random numbers generated must be statistically independent.
3. The sequence of random numbers generated must be reproducible. This allows replication of the simulation experiment.
4. The sequence must be non-repeating for any desired length. Although not theoretically possible, a long repeatability cycle is adequate for practical purposes.

Summary of requirements for a good pseudorandom generator

5. Generation of the random numbers must be fast because in simulation studies a large number of random numbers are required. A slow generator will greatly increase the time and cost of the simulation studies/experiments.
6. The technique used in generating random numbers should require little computer memory.

6.4 Test for Random Numbers

- To check on the desirable properties of random number i.e. uniformity and independence, a number of tests can be performed.
- In testing for uniformity, the hypotheses are as follows:

$$H_0: R_i = U[0,1]$$

$$H_1: R_i \neq U[0,1]$$

- The Null hypothesis, H_0 reads that the number are uniformly distributed on the interval $[0,1]$. Failure to reject null hypothesis means that evidence of non uniformity has not been detected by this test.

6.4 Test for Random Numbers

- In testing for independence, the hypotheses are as follows:

$H_0: R_i = \text{independently generated}$

$H_1: R_i \neq \text{independent generated}$

- This null hypothesis, H_0 reads that the numbers are independent. Failure to reject the null hypothesis means an evidence of dependence has not been detected by this test.
- For each test, a level of significance α must be stated. The level α is the probability of rejecting the null hypothesis when the null hypothesis is true:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$$

6.4 Test for Random Numbers

1. Frequency test

- Uses the Kolmogorov-Smirnov or the chi-square test to compare the distribution of the set of numbers generated to a uniform distribution.

2. Autocorrelation test

- Tests the correlation between numbers and compares the sample correlation to the expected correlation of zero.

6.4 Test for Random Numbers (cont.)

3. Gap test.

- Counts the number of digits that appear between repetitions of a particular digit and then uses the Kolmogorov-Smirnov test to compare with the expected number of gaps.

4. Poker test

- Treats numbers grouped together as a poker hand. Then the hands obtained are compared to what is expected using the chi-square test.

Frequency Test

- Frequency test
 - Frequency test is the test for uniformity.
 - Two different methods of testing are available.
 1. Kolmogorov-Smirnov Test
 2. Chi-Square Test
- Both of these tests measure the degree of agreement between the distribution of sample of generated random numbers and the theoretical uniform distribution.
- Both tests are based on the null hypothesis of no significant difference between the sample distribution and the theoretical distribution.

6.4.1 Kolmogorov-Smirnov Test

Steps Involved:

Step 1: Formulate the hypothesis.

Step 2: rank the data from smallest to largest. Let $R_{(i)}$, denote the smallest observation, so that $R_{(1)} \leq R_{(2)} \leq \dots \leq R_{(N)}$.

Step 3: Compute:

$$D^+ = \max \{ i/N - R_{(i)} \}$$

$$D^- = \max \{ R_{(i)} - i-1/N \}$$

Step 4: Compute $D = \max (D^+, D^-)$

Step 5: Locate in table the critical value, D_α , for the specified significance level α and the given sample size N .

Step 6: if the sample statistic D is greater than the critical value, D_α , the null hypothesis that the data are a sample from a uniform distribution is rejected. If $D \leq D_\alpha$, conclude that no difference has been detected between the true distribution of $\{R_1, R_2, \dots, R_N\}$ and the uniform distribution.

Example

- Suppose that the five numbers 0.44, 0.81, 0.14, 0.05, 0.93 were generated, and it is desired to perform a test for uniformity by using the Kolmogorov-Smirnov test with the level of significance $\alpha = 0.05$.

➤ Step 1: Formulate Hypothesis:

$$H_0: R_i = U[0,1]$$

$$H_1: R_i \neq U[0,1]$$

Example

- Step 2: Perform the calculation as:

$R_{(i)}$	0.05	0.14	0.44	0.81	0.93
i/N	0.20	0.40	0.60	0.80	1.00
$i/N - R_{(i)}$	0.15	0.26	0.16	-	0.07
$R_{(i)} - (i-1)/N$	0.05	-	0.04	0.21	0.13

- The top row in above table lists the number from smallest to largest.
- The computation for $D^+ = 0.26$ and $D^- = 0.21$.
- Therefore, $D = \max(0.26, 0.21) = 0.26$.

Example

- Step 3: Compare the Calculated value with Critical value in table.
 - The critical value of D , obtained from table for $\alpha = 0.05$ and $N = 5$ is 0.56327.
 - Since, the computed value, 0.26 is less than tabulated critical value. 0.56327, the hypothesis that the distribution of the generated numbers is the uniform distribution is not rejected.

Table for Kolmogorov-Smirnov Test

[illegible]

Example

- Perform Kolmogorov-Smirnov test for the following random numbers with level of significance of $\alpha = 0.05$. The critical value for $\alpha = 0.05$ is 0.410 for sample size $N=10$.
- Random numbers are: 0.24, 0.89, 0.11, 0.61, 0.23, 0.86, 0.41, 0.64, 0.50, 0.65

6.4.2 Chi Square Test

- The chi-square test uses the sample statistics:

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed number in the i th class, E_i is the expected number in the i th class, and n is the number of classes. For the uniform distribution E_i , the expected number in each class is given by:

$$E_i = N/n$$

for equally spaced classes, where N is the total number of observations. It can be shown that the sampling distribution of χ_c^2 is approximately the chi-square distribution with $n-1$ degrees of freedom.

Chi-Square Test

Steps Involved.

1. Determine the appropriate test
2. Establish the level of significance: α
3. Formulate the statistical hypothesis
4. Calculate the test statistic
5. Determine the degree of freedom
6. Compare computed test statistic against a tabled/critical value

Chi-Square Test (Example 1)

The two Digit random numbers generated by a multiplicative congruential method are given below. Determine Chi-Square. Is it acceptable at 95% confidence level?

36, 91, 51, 02, 54, 06, 58, 06, 58, 02, 54, 01, 48, 97, 43,
22, 83, 25, 79, 95, 42, 87, 73, 17, 02, 42, 95, 38, 79, 29,
65, 09, 55, 97, 39, 83, 31, 77, 17, 62, 03, 49, 90, 37, 13,
17, 58, 11, 51, 92, 33, 78, 21, 66, 09, 54, 49, 90, 35, 84,
26, 74, 22, 62, 12, 90, 36, 83, 32, 75, 31, 94, 34, 87, 40,
07, 58, 05, 56, 22, 58, 77, 71, 10, 73, 23, 57, 13,
36, 89, 22, 68, 02, 44, 99, 27, 81, 26, 85, 22

Chi-Square Test (Example 1)

➤ Solution,

➤ Step 1: Formulate the Hypothesis:

$$H_0: R_i = U[0,1]$$

$$H_1: R_i \neq U[0,1]$$

➤ Step 2: Calculate the expected frequency as:

$$E_i = N/n$$

$$= 100/10 = 10$$

➤ Step 3: Perform the calculation as:

Class Interval	Frequency (O_i)	$O_i - E_i$	$(O_i - E_i)^2$
$0 < r \leq 10$			
$10 < r \leq 20$			
$20 < r \leq 30$			
$30 < r \leq 40$			
$40 < r \leq 50$			
$50 < r \leq 60$			
$60 < r \leq 70$			
$70 < r \leq 80$			
$80 < r \leq 90$			
$90 < r \leq 100$			

➤ Step 3: Perform the calculation as:

Class Interval	Frequency (O_i)	$O_i - E_i$	$(O_i - E_i)^2$
$0 < r \leq 10$	13	3	9
$10 < r \leq 20$	7	-3	9
$20 < r \leq 30$	12	2	4
$30 < r \leq 40$	13	3	9
$40 < r \leq 50$	7	-3	9
$50 < r \leq 60$	13	3	9
$60 < r \leq 70$	5	-5	25
$70 < r \leq 80$	10	0	0
$80 < r \leq 90$	12	2	4
$90 < r \leq 100$	8	-2	4
			82

$$\text{Chi-Square} = 82/10 = 8.2$$

Chi-Square Test (Example 1)

- Step 4: Compare the Calculated value with the critical value in the table.
- Now, for $9 = (10-1)$ degree of freedom, at 95% confidence level ($\alpha = 0.05$), the acceptable value of Chi-square is 16.919, which is more than 8.2.
- Hence, the given set of random numbers is acceptable, so far as its uniformity in distribution is concerned.

Table for Chi-Square Test

Degrees of Freedom	Probability										
	0.95	0.90	0.80	0.70	0.50	0.30	0.20	0.10	0.05	0.01	0.001
1	0.004	0.02	0.06	0.15	0.46	1.07	1.64	2.71	3.84	6.64	10.83
2	0.10	0.21	0.45	0.71	1.39	2.41	3.22	4.60	5.99	9.21	13.82
3	0.35	0.58	1.01	1.42	2.37	3.66	4.64	6.25	7.82	11.34	16.27
4	0.71	1.06	1.65	2.20	3.36	4.88	5.99	7.78	9.49	13.28	18.47
5	1.14	1.61	2.34	3.00	4.35	6.06	7.29	9.24	11.07	15.09	20.52
6	1.63	2.20	3.07	3.83	5.35	7.23	8.56	10.64	12.59	16.81	22.46
7	2.17	2.83	3.82	4.67	6.35	8.38	9.80	12.02	14.07	18.48	24.32
8	2.73	3.49	4.59	5.53	7.34	9.52	11.03	13.36	15.51	20.09	26.12
9	3.32	4.17	5.38	6.39	8.34	10.66	12.24	14.68	16.92	21.67	27.88
10	3.94	4.86	6.18	7.27	9.34	11.78	13.44	15.99	18.31	23.21	29.59
	Nonsignificant								Significant		

Example 2

- Suppose there are 500 random numbers to be tested for uniformity using Chi-Square test such that the expected frequency in each class intervals is 50. The observed frequency of numbers in each interval is given as:

47, 49, 55, 54, 46, 53, 51, 43, 46 and 54

Perform chi-square test for confidence level of 95%.

Example 2

➤ Step 1: Formulate the Hypothesis:

$$H_0: R_i = U[0,1]$$

$$H_1: R_i \neq U[0,1]$$

➤ Step 2: Calculate the expected frequency as:

$$E_i = N/n$$

$$= 500/10 = 50$$

Example 2

➤ Step 3: Perform the calculation as:

Class	frequency	$(O_i - E_i)$	$(O_i - E_i)^2$
0-50	47	-3	9
50-100	49	-1	1
100-150	55	5	25
150-200	54	4	16
200-250	46	-4	16
250-300	53	3	9
300-350	51	1	1
350-400	43	-7	49
400-450	46	-4	16
450-500	54	4	16
			158

Chi-Square = $\frac{158}{50} = 3.16$

Example 2

- Step 4: Compare the Calculated value with the critical value in the table.
- Now, for $9 = (10-1)$ degree of freedom, at 95% confidence level, the acceptable value of Chi-square is 16.92, which is more than 3.16.
- Hence, the given set of random numbers is acceptable, so far as its uniformity in distribution is concerned.

Example 3

- Using Chi-Square, test the uniformity at 90% for the given random number. Degree of freedom for 9 = 14.684.

20	34	43	42	14	10	33	17	6	11
15	16	4	1	35	22	9	46	37	57
51	49	40	27	59	5	44	19	41	55
53	29	3	31	48	8	56	28	12	7

6.4.3 Gap Test

- The gap test is used to determine the significance of the interval between the recurrences of the same digit.
- The gap of length x occurs between the recurrences of some specified digit.
- Example:
4, 1, 3, 5, 1, 7, 2, 8, 2, 0, 7, 9, 1, 3, 5, 2, 7, 9,
4, 1, 6, 3, 3, 9, 6, 3, 4, 8, 2, 3, 1, ..
- There are a total of 6, 3s therefore number of gaps become 5 and gap length between first two 3s is 10, then 7, 0, 2 and 3 respectively.
- The probability of a particular gap length can be determined by a Bernoulli trail.

$$P(\text{gap of } n) = P(x \neq 3)P(x \neq 3)...P(x \neq 3)P(x = 3)$$

Gap Test

- The theoretical frequency distribution for randomly ordered digits is given by:

$$P(\text{gap} \leq x) = F(x) = 0.1 \sum_{n=0}^x (0.9)^n = 1 - 0.9^{x+1}$$

- The relative (observed) frequency is calculated as:

$$S_N(X) = \text{Frequency of numbers} / \text{No of gaps}$$

Algorithm for GAP Test

Step 1. Specify the cdf(cumulative distribution function) for the theoretical frequency distribution given by Equation (1) based on the selected class interval width.

Step 2. Arrange the observed sample of gaps in a cumulative distribution with these same classes.

Step 3. Find D , the maximum deviation between $F(x)$ and $S_N(x)$ as in Equation $D = \max |F(x) - S_N(x)|$

Step 4. Determine the critical value, D_α from Table(K-S critical value) for the specified value of α and the sample size N .

Step 5. If the calculated value of D is greater than the tabulated value of D_α , the null hypothesis of independence is rejected.

Gap Test (Example)

- Based on the frequency with which the gaps occur, analyze the 110 digits to test whether they are independent. Use $\alpha = 0.05$.

4 1 3 5 1 7 2 8 2 0 7 9 1 3 5 2 7 9 4 1 6 3 3 9
6 3 4 8 2 3 1 9 4 4 6 8 4 1 3 8 9 5 5 7 3 9 5 9
8 5 3 2 2 3 7 4 7 0 3 6 3 5 9 9 5 5 5 0 4 6 8 0
4 7 0 3 3 0 9 5 7 9 5 1 6 6 3 8 8 8 9 2 9 1 8 5
4 4 5 0 2 3 9 7 1 2 0 3 6 3

Gap Test (Example)

- Step 1: Formulate the hypothesis

$H_0: R_i = \text{independently generated}$

$H_1: R_i \neq \text{independent generated}$

- Step 2: Calculate number of Gaps

$$\begin{aligned}\text{Number of gaps} &= \text{Number of data value} - \text{Number of distinct digit} \\ &= 110 - 10 \\ &= 100\end{aligned}$$

Gap Test (Example)

- Step 3: Calculate number of gaps and length of each gaps.

Digits	Length of each gap	No of gaps
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		

Gap Test (Example)

- Step 3: Calculate number of gaps and length of each gaps.

Digits	Length of each gaps	No of gaps
0	47,9,3,2,2,21,6	7
1	2,7,6,10,6,45,9,10	8
2	1,6,12,22,0,38,8,4	8
3	10,7,0,2,3,8,5,5,2,4,1,14,0,9,14,5,1	17
4	17,7,5,0,2,18,12,3,23,0	10
5	10,26,0,3,2,11,2,0,0,12,2,12,2	13
6	3,9,24,9,14,0,22	7
7	4,5,26,10,1,16,6,22	8
8	19,7,3,8,21,16,0,0,4	9
9	5,5,7,8,4,1,14,0,14,2,8,1,9	13

Gap Test (Example)

- Step 4: Calculate Relative Frequency $S_N(X)$ and Theoretical Frequency $F(X)$.

Gap Length	Frequency	Relative Frequency $[S_N(X)] =$ Frequency/No of gaps	Cumulative Relative Frequency $[S_N(X)]$	Theoretical Frequency $[F(X)] = 1 - 0.9^{x+1}$	$ F(X) - S_N(X) $
0-3					
4-7					
8-11					
12-15					
16-19					
20-23					
24-27					
28-31					
32-35					
36-39					
40-43					
44-47					

Gap Test (Example)

- Step 4: Calculate Relative Frequency $S_N(X)$ and Theoretical Frequency $F(X)$.

Gap Length	Frequency	Relative Frequency [$S_N(X)$] = Frequency/No of gaps	Cumulative Relative Frequency [$S_N(X)$]	Theoretical Frequency [$F(X)$] = $1 - 0.9^{x+1}$	$F(X) - S_N(X)$
0-3	35	0.35	0.35	0.3439	0.0061
4-7	22	0.22	0.57	0.5695	0.0005
8-11	17	0.17	0.74	0.7176	0.0224✓
12-15	9	0.09	0.83	0.8147	0.0153
16-19	5	0.05	0.88	0.8784	0.0016
20-23	6	0.06	0.94	0.9202	0.0198
24-27	3	0.03	0.97	0.9497	0.0223
28-31	0	0	0.97	0.9657	0.0043
32-35	0	0	0.97	0.9775	0.0075
36-39	2	0.02	0.99	0.9852	0.0043
40-43	0	0	0.99	0.9903	0.0003
44-47	1	0.01	1.00	0.9936	0.0064

Gap Test (Example)

- Step 5: Calculate the critical value of D for given level of significance against the number of gaps as:

The critical value of D is given by

$$D_{0.05} = 1.36 / \sqrt{100} = 0.136$$

- Step 6: Compare the calculated value with critical value.

Since $D = \max |F(x) - S_N(x)| = 0.0224$ is less than $D_{0.05}$, do not reject the hypothesis of independence on the basis of this test.

6.4.4 Poker's Method

- Poker Test - based on the frequency with which certain digits are repeated.
- Example: 0.255, 0.577, 0.331, 0.414, 0.828, 0.909
- Note: a pair of like digits appear in each number generated.
- In 3-digit numbers, there are only 3 possibilities.
- $P(3 \text{ different digits}) = P(2\text{nd diff. from 1st}) * P(3\text{rd diff. from 1st \& 2nd})$
 $= (0.9) (0.8) = 0.72$
- $P(3 \text{ like digits}) = P(2\text{nd digit same as 1st}) * P(3\text{rd digit same as 2nd})$
 $= (0.1) (0.1) = 0.01$
- $P(\text{exactly one pair}) = P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ digit different as } 1^{\text{st}}) * {}^3C_2$
 $= 3 * 0.1 * 0.9 = 0.27$

OR,

- $P(\text{exactly one pair}) = 1 - 0.72 - 0.01 = 0.27$

Poker Test : (Example)

- A sequence of 1000 three-digit numbers has been generated and an analysis indicates that 680 have three different digits, 289 contain exactly one pair of like digits, and 31 contain three like digits. Based on the poker test, are these numbers independent? Let $\alpha = 0.05$.

➤ Solution,

Step 1: Formulate the hypothesis:

H_0 : R_i = independently generated

H_1 : $R_i \neq$ independently generated

Poker Test : (Example)

Step 2: Calculate the probabilities of possible combinations:

$$\begin{aligned}\text{Case I. } P(3 \text{ different digits}) &= P(2^{\text{nd}} \text{ diff. from } 1^{\text{st}}) * P(3^{\text{rd}} \text{ diff. from } 1^{\text{st}} \& 2^{\text{nd}}) \\ &= (0.9) (0.8) = 0.72\end{aligned}$$

$$\begin{aligned}\text{Case II. } P(3 \text{ like digits}) &= P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ digit same as } 1^{\text{st}}) \\ &= (0.1) (0.1) = 0.01\end{aligned}$$

$$\begin{aligned}\text{Case III. } P(\text{exactly one pair}) &= P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ digit different as } 1^{\text{st}}) * {}^3C_2 \\ &= 3 * 0.1 * 0.9 = 0.27\end{aligned}$$

OR,

$$\text{Case III. } P(\text{exactly one pair}) = 1 - 0.72 - 0.01 = 0.27$$

Poker Test : (Example)

- Step 3: With $N = 1000$, let's summarize the results for Poker's test in the following table:

Combination (i)	Observed Frequency (O _i)	Expected Frequency (E _i)	$(O_i - E_i)^2 / E_i$
Three different digits			
Three like digits			
Exactly one pair			

Poker Test : (Example)

- Step 3: With $N = 1000$, let's summarize the results for Poker's test in the following table:

Combination, i	Observed Frequency, O_i	Expected Frequency, E_i	$\frac{(O_i - E_i)^2}{E_i}$
Three different digits	680	720	2.24
Three like digits	31	10	44.10
Exactly one pair	<u>289</u>	<u>270</u>	<u>1.33</u>
	1000	1000	47.65

The appropriate degrees of freedom are one less than the number of class intervals. Since $\chi^2_{0.05, 2} = 5.99 < 47.65$, the independence of the numbers is rejected on the basis of this test.

Poker Test (Example)

- A sequence of 1000 four-digit numbers has been generated and an analysis indicates the following combinations & frequencies: Use Poker's test to determine if these random numbers are independent, $\alpha = 0.05$.

Combination Distribution (i)	Observed Frequency (O_i)
<i>4 different digits</i>	540
<i>3 like digits</i>	50
<i>4 like digits</i>	20
<i>1 pair</i>	320
<i>2 pairs</i>	70

Poker Test (Example)

➤ Solution,

Step 1: Formulate the hypothesis:

H_0 : R_i = independently generated

H_1 : $R_i \neq$ independently generated

Poker Test (Example)

Step 2: Calculate the probabilities of possible combinations:

In 4-digit numbers, there are only five possibilities- four different, 3 of a kind, 4 of a kind, 1 pair and 2 pairs. Let's calculate the probabilities for each of these cases:

$$\begin{aligned}\text{Case-I: } P(4 \text{ different digits}) &= P(2^{\text{nd}} \text{ diff. from } 1^{\text{st}}) * P(3^{\text{rd}} \text{ diff. from } 1^{\text{st}} \text{ \& } 2^{\text{nd}}) * P(4^{\text{th}} \text{ diff. from } 1^{\text{st}}, 2^{\text{nd}} \text{ \& } 3^{\text{rd}}) \\ &= 0.9 * 0.8 * 0.7 = 0.504\end{aligned}$$

$$\begin{aligned}\text{Case-II: } P(4 \text{ like digits}) &= P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ digit same as } 1^{\text{st}}) * P(4^{\text{th}} \text{ digit same as } 1^{\text{st}}) \\ &= 0.1 * 0.1 * 0.1 = 0.001\end{aligned}$$

Poker Test (Example)

Case-III: $P(3 \text{ like digits}) = P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ digit same as } 1^{\text{st}}) * P(4^{\text{th}} \text{ digit diff. from } 1^{\text{st}})$
* No. of possible combinations for 3 like digits from 4 digits
 $= 0.1 * 0.1 * 0.9 * {}^4C_3 = 0.1 * 0.1 * 0.9 * 4 = 0.036$

Case-IV: $P(1 \text{ pair}) = P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}} \text{ in the pair}) * P(3^{\text{rd}} \text{ digit diff. from } 1^{\text{st}}) * P(4^{\text{th}} \text{ digit diff. from } 1^{\text{st}} \& 3^{\text{rd}})$
* No. of possible combinations for a pair from 4 digits
 $= 0.1 * 0.9 * 0.8 * {}^4C_2 = 0.1 * 0.9 * 0.8 * 6 = 0.432$

Case-V: $P(2 \text{ pairs}) = P(2^{\text{nd}} \text{ digit same as } 1^{\text{st}} \text{ in the pair}) * P(3^{\text{rd}} \text{ digit diff. from } 1^{\text{st}}) * P(4^{\text{th}} \text{ digit same as } 3^{\text{rd}})$
* No. of possible combinations for two pairs from 4 digits
 $= 0.1 * 0.9 * 0.1 * (\frac{1}{2} * {}^4C_2) = 0.1 * 0.9 * 0.1 * 3 = 0.027$

OR,

Case-V: $P(2 \text{ pairs}) = 1 - P(4 \text{ different digits}) - P(3 \text{ like digits}) - P(4 \text{ like digits}) - P(1 \text{ pair})$
 $= 1 - 0.504 - 0.036 - 0.001 - 0.432 = 0.027$

Poker Test (Example)

- Step 3: With $N = 1000$, let's summarize the results for Poker's test in the following table:

Combination Distribution (i)	Observed Frequency O_i	Expected Frequency $E_i = \text{Prob.} * N$	$(O_i - E_i)^2 / E_i$
<i>4 different digits</i>	540	504	2.571
<i>3 like digits</i>	50	36	5.44
<i>4 like digits</i>	20	1	361
<i>1 pair</i>	320	432	29.037
<i>2 pairs</i>	70	27	68.48
	1000	1000	466.528

Poker Test (Example)

Step 4: Compare the calculated value with critical value in the table.

Calculated χ^2 value = 466.528

Degree of freedom = $n - 1 = 5 - 1 = 4$

At $\alpha = 0.05$, acceptable value of χ^2 from table = 9.49

Since, the calculated value of χ^2 is greater than the acceptable value at four degree of freedom, the independence of the random numbers is rejected on the basis of this test.

Poker Test (Example)

- A sequence of 10,000 five-digit random number has been generated, and an analysis of numbers indicate that there are 3075 numbers having five different digits, 4935 having a pair, 1135 having two pairs 695 having three of a kind, 105 having full house (3 of a kind and a pair), 54 having four of a kind and one having all five of a kind. Use poker test to determine if these random numbers are independent, at $\alpha = 0.01$.

Poker Test (Example)

➤ Solution,

Step 1: Formulate the hypothesis:

H_0 : R_i = independently generated

H_1 : $R_i \neq$ independently generated

Poker Test (Example)

- Step 2: Calculate the probabilities of possible combinations:
- In a five digit number there are seven possibilities: 5 different digits, 1 pair, two pairs, three of a kinds, full houses (3 of a kind and a pair), four of a kinds and five of a kinds. Let's calculate the probabilities for each of these cases:

Case-I: $P(5 \text{ different digits}) = P(2^{\text{nd}} \text{ diff. from } 1^{\text{st}}) * P(3^{\text{rd}} \text{ diff. from } 1^{\text{st}} \& 2^{\text{nd}}) * P(4^{\text{th}} \text{ diff. from } 1^{\text{st}}, 2^{\text{nd}} \& 3^{\text{rd}}) * P(5^{\text{th}} \text{ diff. from } 1^{\text{st}}, 2^{\text{nd}}, 3^{\text{rd}} \& 4^{\text{th}})$
 $= 0.9 * 0.8 * 0.7 * 0.6 = 0.3024$

Case-II: $P(5 \text{ same digits}) = P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ same as } 2^{\text{nd}}) * P(4^{\text{th}} \text{ same as } 3^{\text{rd}}) * P(5^{\text{th}} \text{ same as } 4^{\text{th}})$
 $= 0.1 * 0.1 * 0.1 * 0.1 = 0.0001$

Poker Test (Example)

$$\begin{aligned}\text{Case-III: } P(4 \text{ same digits}) &= P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ same as } 2^{\text{nd}}) * \\ &P(4^{\text{th}} \text{ same as } 3^{\text{rd}}) * P(5^{\text{th}} \text{ diff. from } 1^{\text{st}}) * {}^5C_4 \\ &= 0.1 * 0.1 * 0.1 * 0.9 * 5 = 0.0045\end{aligned}$$

$$\begin{aligned}\text{Case-IV: } P(\text{Full Houses}) &= P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ same as } 2^{\text{nd}}) * P(4^{\text{th}} \\ &\text{diff from } 1^{\text{st}}) * P(5^{\text{th}} \text{ same as } 4^{\text{th}}) * {}^5C_2 \\ &= 0.1 * 0.1 * 0.9 * 0.1 * 10 = 0.009\end{aligned}$$

Poker Test (Example)

$$\begin{aligned}\text{Case-V: } P(3 \text{ same digits}) &= P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ same as } 2^{\text{nd}}) * P(4^{\text{th}} \text{ different from } 1^{\text{st}}) * P(5^{\text{th}} \text{ diff. from } 1^{\text{st}} \text{ and } 4^{\text{th}}) * {}^5C_3 \\ &= 0.1 * 0.1 * 0.9 * 0.8 * 10 = 0.072\end{aligned}$$

$$\begin{aligned}\text{Case-VI: } P(\text{One Pairs}) &= P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ differ from } 1^{\text{st}}) * P(4^{\text{th}} \text{ diff from } 1^{\text{st}} \text{ and } 3^{\text{rd}}) * P(5^{\text{th}} \text{ diff. from } 1^{\text{st}}, 3^{\text{rd}} \text{ and } 4^{\text{th}}) * {}^5C_3 \\ &= 0.1 * 0.9 * 0.8 * 0.7 * 10 = 0.504\end{aligned}$$

$$\begin{aligned}\text{Case-VII: } P(\text{Two Pairs}) &= P(2^{\text{nd}} \text{ same as } 1^{\text{st}}) * P(3^{\text{rd}} \text{ differ from } 1^{\text{st}}) * P(4^{\text{th}} \text{ digit same as } 3^{\text{rd}}) * P(5^{\text{th}} \text{ diff. from } 1^{\text{st}} \text{ and } 3^{\text{rd}}) * \frac{1}{2} * {}^4C_2 * {}^5C_1 \\ &= 0.1 * 0.9 * 0.1 * 0.8 * 3 * 5 = 0.108\end{aligned}$$

OR,

$$\begin{aligned}\text{Case-VII: } P(\text{Two Pairs}) &= 1 - 0.3024 - 0.0001 - 0.0045 - 0.009 - 0.072 - 0.504 \\ &= 0.108\end{aligned}$$

Poker Test (Example)

- Step 3: With $N = 10,000$, let's summarize the results for Poker's test in the following table:

Combination Distribution (i)	Observed Frequency O_i	Expected Frequency $E_i = \text{Prob.} * N$	$(O_i - E_i)^2 / E_i$
Five Different Digits	3075	3024	0.8601
One Pair	4935	5040	2.1875
Two Pairs	1135	1080	2.8009
Three of a kind	695	720	0.868
Full Houses	105	90	2.5
Four of a kind	54	45	1.8
Five of a kind	01	01	0.0
	10,000	10,000	11.0165

Poker Test (Example)

- Step 4: Compare the calculated value with the critical value in the table.
- Calculated χ^2 value = 11.0165

Degree of freedom = $n - 1 = 7 - 1 = 6$

At $\alpha = 0.01$, acceptable value of χ^2 from table = 16.8.

Since, the calculated value of χ^2 is less than the acceptable value at six degree of freedom, the independence of the random numbers is accepted on the basis of this test.

What do you mean by poker test? A sequence of 10,000 five-digit numbers has been generated and an analysis shows following combinations and frequencies. Based on Poker test, test these numbers are independent or not for $\alpha = 0.05$. (Use 12.59 as Tabular Value).

[2+8]

Combinations	Observed Frequency
All different Digits	3054
One Pair	5020
Two Pairs	1073
Three of a kind	710
Full House	95
Four of a kind	44
Five of a kind	4

6.4.5 Testing for auto correlation

- Autocorrelation measures the relationship between numbers in a random number sequence.
- It determines whether numbers separated by a fixed interval (lag) are independent.
- A good random number generator should produce no autocorrelation, meaning each number is independent of the others.

6.4.5 Testing for auto correlation

0.12	0.01	0.23	0.28	0.89
0.31	0.64	0.28	0.83	0.93
0.41	0.60	0.27	0.75	0.88

Observe every 5th number: 0.89, 0.93, 0.88

These values are consistently high, suggesting a possible dependence between numbers separated by 5 positions (lag = 5).

6.4.5 Testing for auto correlation

1. Formulate the hypothesis

$$H_0 : \rho_{im} = 0$$

$$H_1 : \rho_{im} \neq 0$$

2. Determine the largest integer M such that:

$$i + (M+1)m \leq N$$

3. Estimate the autocorrelation:

$$\hat{\rho}_{im} = \frac{1}{M+1} \sum_{k=0}^M R_{i+km} R_{i+(k+1)m} - 0.25$$

4. Compute theoretical standard deviation of auto correlation

$$\sigma_{\hat{\rho}} = \frac{\sqrt{13M+7}}{12(M+1)}$$

6.4.5 Testing for auto correlation

5. Calculate test statistics

$$Z_0 = \frac{\hat{\rho}_{im}}{\sigma_{\hat{\rho}_{im}}}$$

6. Determine the critical value $z_{\alpha/2}$ for the chosen significance level α .

7. Decision Rule

- If $|Z_0| > z_{\alpha/2}$, **Reject H0** (Autocorrelation exists.)
- If $|Z_0| \leq z_{\alpha/2}$, **Fail to reject H0** (No evidence of autocorrelation.)

6.4.5 Testing for auto correlation (Example)

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

Test for whether the 3rd, 8th, 13th, and so on, numbers in the sequence at the beginning of this section are autocorrelated. Assume a significance level of $\alpha=0.05$ with $z_{\alpha/2}=1.96$.

6.4.5 Testing for auto correlation

Solution:

Step 1: Formulate the hypothesis

$$H_0 : \rho_{im} = 0$$

$$H_1 : \rho_{im} \neq 0$$

Step 2: Determine M

Here, $i=3$, $m=8-3=5$, $N=30$

$$i+(M+1)m \leq N$$

$$3+(M+1)5 \leq 30$$

$$M+1 \leq 5.4$$

$$M \leq 4.4$$

Since M must be an integer, the largest possible value is 4. So, $M=4$

6.4.5 Testing for auto correlation

Step 3: Estimate autocorrelation

$$\hat{\rho}_{im} = \frac{1}{M+1} \sum_{k=0}^M R_{i+km} R_{i+(k+1)m} - 0.25$$

$$\hat{\rho}_{3,5} = \frac{1}{4+1} \sum_{k=0}^4 R_{3+5k} R_{3+5(k+1)} - 0.25$$

$$= \frac{1}{5} [R_3 R_8 + R_8 R_{13} + R_{13} R_{18} + R_{18} R_{23} + R_{23} R_{28}] - 0.25$$

$$= \frac{1}{5} [(0.23 \times 0.28) + (0.28 \times 0.33) + (0.33 \times 0.27) + (0.27 \times 0.05) + (0.05 \times 0.36)] - 0.25$$

$$= -0.19452$$

6.4.5 Testing for auto correlation

Step 4: Compute theoretical standard deviation of auto correlation

$$\sigma_{\hat{\rho}} = \frac{\sqrt{13M + 7}}{12(M + 1)}$$

$$\sigma_{\hat{\rho}_{35}} = \frac{\sqrt{13 \times 4 + 7}}{12 \times (4 + 1)} = 0.1280$$

Step 5: Calculate Z_0 .

$$Z_0 = \frac{\hat{\rho}_{im}}{\sigma_{\hat{\rho}_{im}}} = \frac{-0.1945}{0.1280} = -1.519$$

6.4.5 Testing for auto correlation

Step 6: Compare the calculated value with the critical value in the table.

- Calculated Z_0 value = -1.516

At $\alpha = 0.05$, $z_{\alpha/2} = 1.96$

Since $Z_0 < z_{\alpha/2}$, the null hypothesis cannot be rejected.

6.4.5 Testing for auto correlation

The sequence of random numbers 0.23, 0.45, 0.67, 0.12, 0.89, 0.34, 0.56, 0.78, 0.19, 0.41, 0.63, 0.08, 0.85, 0.29, 0.51, 0.73, 0.16, 0.94, 0.37, 0.59 has been generated. Test for the independence among the numbers in the sequence starting with index (i) =2 and lag (m) =3 using the autocorrelation test. Assume a significance level of $\alpha=0.05$ with $z_{\alpha/2}=1.96$.

6.5 Generating discrete distributions

- Generally, in real world implementation there is need for non-uniform random number generation i.e., the distribution is not uniform, so that a different probability is associated with each output.
- Suppose, for example , it is necessary to generate a random variable representing the number of items bought by a shopper at store , where the probability function is the discrete distribution, given in table 1 below.

6.5 Generating discrete distributions

Table 1: Number of Items Bought by customers

Number of Items X_i	Number of Customers N_i	Probability Distribution $P(X_i)$	Cumulative Distribution $P(X_i)$
1	25	0.10	0.10
2	128	0.51	0.61
3	47	0.19	0.80
4	38	0.15	0.95
5	12	0.05	1.00

6.5 Generating discrete distributions

- Then again a table is formed to list the number of items, X , and the cumulative probability, Y , as shown in table 2.

Table 2: Generating a Non-Uniform Discrete Distribution

No of Items X	Probability $P(X)$	Cumulative Probability y
0	0	0
1	0.10	0.10
2	0.51	0.61
3	0.19	0.80
4	0.15	0.95
5	0.05	1.00

6.5 Generating discrete distributions

- Now, taking the output of a uniform random number generator, U , the value is compared with the values of Y .
- If the value falls in an interval $y_i < U \leq y_{i+1}$ ($i=0,1,\dots,4$), the corresponding value of x_{i+1} is taken as desired output.
- It is not necessary that the intervals be in any particular order.
- A computer routine will usually search the table from the first entry .
- The amount of searching can be minimized by selecting the intervals in decreasing order of probability.

6.5 Generating discrete distributions

- Arranged as a table for a computer routine, the data of Table 2, would then appear as shown in table 3,

Table 3: Non-Uniform Discrete Distribution Data
Arranged for a Computer Program

Cumulative Probability U	No. of Items X
0.51	2
0.70	3
0.85	4
0.95	1
1.00	5

6.5 Generating discrete distributions

- With this arrangement , 51% of the searches will only need to go to the first entry , 70% to the first or second and so on.
- With the original ordering, only 10% are satisfied with the first entry and only 61% with the first two.
- The input of table 3 is being denoted by U to indicate a random number between 0 and 1 however the output continues to be denoted by X .

6.6 Inversion, Rejection, Composition and Convolution

Inversion or Inverse Transformation Method

If U_i ($i = 1, 2, 3, 4, \dots$) are independent variables, uniformly distributed over the interval 0 to 1, and $F^{-1}(x)$ is the inverse of the cumulative distribution function for random variable x , then the random variables defined by $x_i = F^{-1}(U_i)$ are the random sample of x .

Inversion or Inverse Transformation Method

- Example: If the desired probability distribution is:

CDF

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

now, find $U(0,1)$

then,

$$F(x) = R$$

or

$$1 - e^{-\lambda x} = R$$

$$e^{-\lambda x} = 1 - R$$

$$-\lambda x = \ln(1 - R)$$

$$x = \left(-\frac{1}{\lambda} \ln(R) \right)$$

Rejection

- The rejection method is applied when the probability density function, $f(x)$, has a lower and upper limit to its range, a and b , respectively, and an upper bound c .
- The method can be specified as follows:
 - Compute the values of two, independent uniformly distributed variates U_1 and U_2
 - Compute $X_0 = a + U_1(b-a)$.
 - Compute $Y_0 = cU_2$
 - If $Y_0 \leq f(X_0)$ accept X_0 as the desired output; otherwise repeat the process with two new uniform variates.

Rejection

- This method is closely related to the process of evaluating an integral using the Monte Carlo technique.
- The probability density function is enclosed in a rectangle with sides of lengths $b-a$ and c .
- The first three steps of rejection method creates just a random point and the last step relates the point to the curve of the probability density function.

Composition

- Sometimes the random variables X of interest involves the sum of $n > 1$ independent random variables:

$$X = Y_1 + Y_2 + Y_3 + \dots + Y_n .$$

- To generate a value for X , we can generate a value for each of the random variables Y_1, Y_2, Y_3, Y_n and add them together. This is called composition.
- Composition can also be used to generate random numbers that are approximately normally distributed.

Composition

- The normal distributed is one of the most important and frequently used continuous distributions.
- The notion $N(\mu, \sigma^2)$ refers to the normal distribution with mean μ and variance σ^2 .
- The central limit theorem in probability says that if Y_1, Y_2, Y_3, Y_n are independent and identically distributed random variables with mean μ and positive variance σ^2 , then is random variable

$$Z = \frac{\sqrt{n} \left(\left(\frac{x}{n} \right) - \mu \right)}{\sigma}$$

Convolution Method

- The probability distribution of a sum of two or more independent random variables is called a convolution of the distributions of the original variables.
- The convolution method thus refers to adding together two or more random variables to obtain a new random variable with the desired distribution.

Convolution Method

- Technique can be used for all random variables X that can be expressed as the sum of n random variables $X = Y_1 + Y_2 + Y_3 + \dots + Y_n$
- In this case, one can generate a random variate X by generating n random variates, one from each of the Y_i , and summing them.
- Examples of random variables:
 - Sum of n Bernoulli random variables is a binomial random variable.
 - Sum of n exponential random variables is an n - Erlang random variable.

References

- Jerry Banks, John S. Carson, II Barry L. Nelson , David M. Nocol, “Discrete Event system simulation”
- Geoffrey Gordon, System Simulation